

EA-0032 - AQELYN Engineering Archive

AQELYN - EA-0032 Engineering Archive

IS-032 - AQELYN Secrets Security & Cryptographic Asset Intelligence Engine

****Archive ID:**** EA-0032

****Implementation Specification:**** IS-032

****Component:**** AQELYN Secrets Security & Cryptographic Asset Intelligence Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Predecessor Archives:**** EA-0001 through EA-0031

****Next Specification:**** IS-033 - AQELYN Identity Security Posture Management (ISPM) Intelligence Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0032 |

| Specification | IS-032 - AQELYN Secrets Security & Cryptographic Asset Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0032.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-032 complete; EA-0032 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

| EA-0004 | IS-004 | AQELYN Evidence Engine |

- | EA-0005 | IS-005 | AQELYN Knowledge Graph |
- | EA-0006 | IS-006 | AQELYN Trust Engine |
- | EA-0007 | IS-007 | AQELYN Mission Engine |
- | EA-0008 | IS-008 | AQELYN Workflow Engine |
- | EA-0009 | IS-009 | AQELYN Policy Engine |
- | EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |
- | EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |
- | EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |
- | EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |
- | EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |
- | EA-0015 | IS-015 | AQELYN Security Operations Engine |
- | EA-0016 | IS-016 | AQELYN Digital Forensics Engine |
- | EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |
- | EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |
- | EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |
- | EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |
- | EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |
- | EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |
- | EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |
- | EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |
- | EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |
- | EA-0026 | IS-026 | AQELYN Configuration Compliance & Drift Intelligence Engine |
- | EA-0027 | IS-027 | AQELYN Identity Threat Detection & Behavioral Analytics Engine |
- | EA-0028 | IS-028 | AQELYN Cloud Security Posture Management Intelligence Engine |
- | EA-0029 | IS-029 | AQELYN SaaS Security Posture Management Intelligence Engine |
- | EA-0030 | IS-030 | AQELYN Software Supply Chain Security & SBOM Intelligence Engine |
- | EA-0031 | IS-031 | AQELYN Data Security Posture Management Intelligence Engine |
- | EA-0032 | IS-032 | AQELYN Secrets Security & Cryptographic Asset Intelligence Engine |

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

2. IS-032 Specification Identity

Specification ID: IS-032

Name: AQELYN Secrets Security & Cryptographic Asset Intelligence Engine

Engineering Archive Target: EA-0032

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: Complete

Predecessor: IS-031 - AQELYN Data Security Posture Management Intelligence Engine

3. Purpose

The AQELYN Secrets Security & Cryptographic Asset Intelligence Engine provides continuous discovery, inventory, governance, risk assessment, and lifecycle management of enterprise secrets and cryptographic assets across cloud, on-premises, SaaS, containerized, and hybrid environments.

The engine continuously discovers secrets, encryption keys, certificates, tokens, API credentials, SSH keys, and cryptographic material; evaluates their exposure and lifecycle; validates cryptographic policy compliance; and correlates cryptographic risk using explainable AI and policy-driven analytics.

It answers:

- Where are enterprise secrets located?
- Which cryptographic assets are expired or vulnerable?
- Which certificates require rotation?
- Which secrets are exposed publicly?
- Which encryption keys violate enterprise policy?
- Can every cryptographic assessment be explained and audited?

4. Mission

The engine shall provide:

- Continuous secret discovery
- Cryptographic asset inventory
- Certificate lifecycle management
- Encryption key governance
- Secret exposure analysis
- Cryptographic policy validation
- Mission-aware cryptographic risk reporting
- Executive cryptographic posture summaries
- Continuous reassessment
- Enterprise cryptographic governance

5. Scope

5.1 In Scope

- API keys
- Secrets
- Passwords
- Certificates
- TLS/SSL certificates
- SSH keys
- Encryption keys
- HSM-managed keys
- KMS-managed keys
- Cloud secrets managers
- Container secrets
- Vault systems
- Cryptographic tokens

5.2 Out of Scope

- Application source code development
- PKI infrastructure implementation
- Hardware manufacturing
- Network encryption protocols
- End-user password management

6. Dependencies

IS-032 depends on:

```
IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
IS-016 Digital Forensics Engine
IS-017 Threat Detection & Analytics Engine
IS-018 Automated Response & Orchestration Engine
IS-019 Security Data Lake & Telemetry Platform
IS-020 AI Decision Intelligence Engine
IS-021 Predictive Analytics & Forecasting Engine
IS-022 Executive Intelligence & Strategic Reporting Engine
IS-023 Threat Exposure & Attack Surface Management Engine
IS-024 Vulnerability Intelligence & Prioritization Engine
IS-025 Cyber Asset Discovery & Inventory Intelligence Engine
IS-026 Configuration Compliance & Drift Intelligence Engine
IS-027 Identity Threat Detection & Behavioral Analytics Engine
IS-028 Cloud Security Posture Management Intelligence Engine
IS-029 SaaS Security Posture Management Intelligence Engine
IS-030 Software Supply Chain Security & SBOM Intelligence Engine
IS-031 Data Security Posture Management Intelligence Engine
```

7. High-Level Architecture

```
AQELYN Secrets Security & Cryptographic Asset Intelligence Engine
■
■■■ Secret Discovery Engine
■■■ Cryptographic Asset Inventory Engine
■■■ Certificate Lifecycle Engine
■■■ Key Management Intelligence Engine
■■■ Secret Exposure Analysis Engine
■■■ Cryptographic Compliance Engine
■■■ Cryptographic Risk Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-032-001 - Secret Discovery

The engine shall continuously discover:

```
Passwords
API keys
Tokens
Secrets
Certificates
SSH keys
Encryption keys
Vault-managed credentials
```

FR-032-002 - Cryptographic Asset Inventory

Continuously inventory:

- TLS certificates
- Code-signing certificates
- Encryption keys
- KMS keys
- HSM keys
- Certificate authorities
- Cryptographic algorithms
- Key ownership

FR-032-003 - Cryptographic Risk Detection

Detect:

- Expired certificates
- Weak encryption
- Hardcoded secrets
- Public secret exposure
- Key reuse
- Unapproved algorithms
- Missing rotation
- Policy violations

FR-032-004 - Explainable Cryptographic Intelligence

Every assessment shall include:

- Evidence references
- Confidence indicators
- Policy rationale
- Historical lifecycle
- Risk explanation
- Recommended remediation

FR-032-005 - Governance

Support:

- Certificate governance
- Key governance
- Secret governance
- Approval workflows
- Policy validation
- Auditability
- Executive review

FR-032-006 - Event Publication

Publish standardized events:

- secret.discovered
- certificate.expiring
- certificate.expired
- key.rotation.required
- secret.exposure.detected
- cryptography.policy.violation
- cryptography.risk.updated
- cryptography.remediation.recommended

9. Non-Functional Requirements

The engine shall provide:

Continuous assessment
Enterprise scalability
Low-latency cryptographic analysis
Explainability
Auditability
Repository stability
Backward compatibility

10. Core Cryptographic Assessment Lifecycle

```

graph TD
    A[Secret or Key Discovered] --> B[Inventory Updated]
    B --> C[Exposure Assessment]
    C --> D[Compliance Validation]
    D --> E[Risk Assessment]
    E --> F[Rotation / Remediation]
    F --> G[Continuous Monitoring]
  
```

11. Internal Component Architecture

The AQELYN Secrets Security & Cryptographic Asset Intelligence Engine is implemented as a modular enterprise cryptographic security platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Identity Governance Engine, Cloud Security Posture Management Engine, Data Security Posture Management Engine, Risk Intelligence Engine, AI Decision Intelligence Engine, Executive Intelligence Engine, and Security Operations Engine.

```

graph TD
    Root[AQELYN Secrets Security & Cryptographic Asset Intelligence Engine]
    Root --- SDE[Secret Discovery Engine]
    Root --- CAIE[Cryptographic Asset Inventory Engine]
    Root --- CLE[Certificate Lifecycle Engine]
    Root --- KMIIE[Key Management Intelligence Engine]
    Root --- SEAE[Secret Exposure Analysis Engine]
    Root --- CCE[Cryptographic Compliance Engine]
    Root --- CRE[Cryptographic Risk Engine]
    Root --- KGC[Knowledge Graph Connector]
    Root --- SDLC[Security Data Lake Connector]
    Root --- ADC[AI Decision Connector]
    Root --- ERC[Executive Reporting Connector]
    Root --- EP[Event Publisher]
  
```

12. Component Specifications

12.1 Secret Discovery Engine

Continuously discovers enterprise secrets and credentials.

Capabilities:

API key discovery
Password discovery
Secret scanning
Vault inventory
Token discovery
Credential cataloging

12.2 Cryptographic Asset Inventory Engine

Maintains a complete inventory of cryptographic assets.

Supports:

- TLS certificates
- SSH keys
- Code-signing certificates
- Encryption keys
- KMS-managed keys
- HSM-managed keys

12.3 Certificate Lifecycle Engine

Monitors certificate validity and lifecycle.

Produces:

- Expiration tracking
- Renewal recommendations
- Certificate ownership
- Chain validation
- Lifecycle history

12.4 Key Management Intelligence Engine

Evaluates encryption key lifecycle and governance.

Produces:

- Rotation status
- Key age analysis
- Key ownership
- Algorithm validation
- Key usage analytics

12.5 Secret Exposure Analysis Engine

Analyzes exposure and misuse of secrets.

Produces:

- Public exposure detection
- Hardcoded secret detection
- Repository exposure
- Container secret analysis
- Secret leakage indicators

12.6 Cryptographic Compliance Engine

Validates cryptographic assets against enterprise policy.

Supports:

- Policy compliance scoring
- Algorithm validation
- Certificate policy checks
- Rotation compliance
- Audit evidence

12.7 Cryptographic Risk Engine

Calculates enterprise cryptographic security risk.

Produces:

```
Cryptographic risk score  
Mission impact  
Business impact  
Threat likelihood  
Remediation priority
```

13. Universal Object Model Extensions

13.1 SecretAsset

```
SecretAsset:  
secret_id  
secret_type  
owner  
storage_location
```

13.2 CryptographicKey

```
CryptographicKey:  
key_id  
algorithm  
strength  
rotation_date
```

13.3 CertificateAsset

```
CertificateAsset:  
certificate_id  
issuer  
expiration_date  
validation_status
```

13.4 CryptographicExposure

```
CryptographicExposure:  
exposure_id  
exposure_type  
severity  
detected_at
```

14. Knowledge Graph Integration

Relationships:

```
Application  
↓  
contains  
↓  
SecretAsset  
  
SecretAsset  
↓  
protected_by  
↓  
CryptographicKey  
  
CryptographicKey  
↓  
validated_by  
↓  
CertificateAsset  
  
CertificateAsset  
↓  
associated_with  
↓  
CryptographicExposure
```


15. Event Bus Integration

15.1 Discovery Events

```
secret.discovered  
secret.updated  
secret.removed
```

15.2 Certificate Events

```
certificate.expiring  
certificate.expired  
certificate.renewed
```

15.3 Key Events

```
key.rotation.required  
key.rotated  
key.policy.violation
```

15.4 Governance Events

```
secret.exposure.detected  
cryptography.policy.violation  
cryptography.risk.updated  
cryptography.remediation.recommended
```

16. Security Data Lake Integration

Consumes:

```
Secret inventories  
Certificate metadata  
Key metadata  
Vault telemetry  
Access logs  
Repository scan results
```

17. AI Decision Intelligence Integration

Consumes:

```
Cryptographic risk scores  
Exposure intelligence  
Compliance trends  
Remediation recommendations  
Confidence indicators
```

18. Risk & Threat Integration

Consumes:

```
Threat intelligence  
Business criticality  
Mission impact  
Identity risk
```

Software supply chain intelligence
Data security posture intelligence

19. Compliance Integration

Supports:

Certificate governance
Key governance
Secret governance
Policy validation
Audit evidence
Executive reporting

20. Public APIs

20.1 Secret Inventory API

GET /secrets
GET /secrets/{id}

20.2 Certificate API

GET /certificates
GET /certificates/{id}

20.3 Cryptographic Key API

GET /keys
GET /keys/{id}

20.4 Exposure API

GET /cryptography/exposures
GET /cryptography/exposures/{id}

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/  
■■■ src/  
■ ■■■ cryptographic_security/  
■■■ tests/  
■ ■■■ cryptographic_security/  
■■■ docs/  
■ ■■■ cryptographic_security/  
■■■ api/  
■ ■■■ cryptographic_security/  
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Secrets Security & Cryptographic Asset Intelligence Engine is the trusted subsystem responsible for continuously discovering enterprise secrets, managing cryptographic assets, monitoring certificate lifecycles, validating encryption policies, detecting exposed credentials, and governing enterprise cryptographic posture.

Every cryptographic assessment shall be:

- Explainable
- Evidence-backed
- Policy-governed
- Cryptographically aware
- Risk-aware
- Mission-aware
- Fully auditable
- Continuously reassessed

22.1 Security Principles

- Zero Trust
- Defense in Depth
- Least Privilege
- Continuous Cryptographic Assessment
- Secure by Design
- Policy Enforcement
- Explainable Cryptographic Intelligence
- Continuous Monitoring

22.2 Authorization Model

Supported operational roles:

- Secrets Administrator
- PKI Administrator
- Cryptographic Security Engineer
- Security Administrator
- SOC Analyst
- Compliance Officer
- Mission Owner
- Executive Reviewer

All cryptographic assessments, secret rotation decisions, and certificate governance actions shall be governed through the AQELYN Policy Engine.

22.3 Secret Assessment Integrity

Secret and cryptographic assessments shall maintain:

- Unique assessment identifier
- Secret or cryptographic asset reference
- Ownership reference
- Evidence references
- Risk score
- Confidence indicators
- Exposure context
- Remediation guidance
- Audit trail

Assessment history shall be append-only.

22.4 Key and Certificate Integrity

Cryptographic asset records shall maintain:

- Asset identifier
- Asset type
- Algorithm
- Strength
- Issuer or authority
- Owner
- Rotation state

Expiration state
Validation state
Evidence references
Audit record

No cryptographic asset shall be considered authoritative without ownership, validation metadata, and policy references.

23. Cryptographic Lifecycle

23.1 Discovery Lifecycle

Secret / Certificate / Key Discovered
↓
Inventory Updated
↓
Exposure Assessment
↓
Compliance Validation
↓
Risk Assessment
↓
Continuous Monitoring

23.2 Rotation Lifecycle

Rotation Required
↓
Risk Prioritized
↓
Approval Granted
↓
Rotation Executed
↓
Validation Completed
↓
Audit Recorded

23.3 Audit Lifecycle

Assessment Created
↓
Evidence Linked
↓
Policy Validated
↓
Audit Stored

24. Continuous Cryptographic Operations

The engine continuously evaluates:

Enterprise secrets
TLS certificates
SSH keys
Encryption keys
Vault systems
Secrets managers
HSM assets
KMS assets
Certificate authorities
Cryptographic algorithms

25. Performance Requirements

The engine shall support:

- Continuous assessment
- Low-latency cryptographic analysis
- Enterprise-scale inventories
- Concurrent certificate validation
- High availability
- Continuous operation

26. Scalability Requirements

The engine shall scale to support:

- Global enterprises
- Millions of secrets
- Millions of certificates
- Large PKI environments
- Distributed vault systems
- Long-term cryptographic history

27. Audit Requirements

Every cryptographic operation shall generate immutable audit records.

Audit events include:

- Secret discovered
- Certificate renewed
- Certificate expired
- Key rotated
- Policy violation identified
- Risk updated
- Exposure detected
- Remediation completed

28. Failure Handling

28.1 Discovery Failure

- Discovery retried
- Failure recorded
- Administrator notified

28.2 Certificate Validation Failure

- Validation retried
- Previous assessment retained
- Audit generated

28.3 Secret Exposure Analysis Failure

- Analysis repeated
- Manual review initiated
- Audit recorded

28.4 Policy Failure

- Rotation blocked
- Policy violation recorded
- Manual approval required

29. Testing Strategy

29.1 Unit Testing

Validate:

```
Secret Discovery Engine
Cryptographic Asset Inventory Engine
Certificate Lifecycle Engine
Key Management Intelligence Engine
Secret Exposure Analysis Engine
Cryptographic Compliance Engine
Cryptographic Risk Engine
```

29.2 Integration Testing

Verify interaction with:

```
Kernel
Knowledge Graph
Security Data Lake
Identity Governance Engine
Cloud Security Posture Management Engine
DSPM Intelligence Engine
Risk Intelligence Engine
AI Decision Engine
Executive Reporting Engine
Security Operations Engine
```

29.3 System Testing

Validate:

```
Secret discovery
Certificate lifecycle
Key governance
Exposure detection
Risk scoring
Audit generation
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Cryptographic assessment integrity
```

29.5 Regression Testing

Verify IS-001 through IS-031 remain unaffected.

30. Acceptance Criteria

IS-032 is complete when:

```
Secret Discovery Engine implemented
Cryptographic Asset Inventory Engine implemented
```

```
Certificate Lifecycle Engine implemented
Key Management Intelligence Engine implemented
Secret Exposure Analysis Engine implemented
Cryptographic Compliance Engine implemented
Cryptographic Risk Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/cryptographic_security/
■■■ tests/cryptographic_security/
■■■ docs/cryptographic_security/
■■■ api/cryptographic_security/
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-032 introduces the AQELYN Secrets Security & Cryptographic Asset Intelligence Engine, providing enterprise-scale discovery, governance, lifecycle management, compliance validation, exposure analysis, explainable cryptographic risk scoring, and continuous governance of secrets and cryptographic assets.

Major capabilities include:

```
Secret Discovery
Cryptographic Asset Inventory
Certificate Lifecycle Management
Key Management Intelligence
Secret Exposure Analysis
Cryptographic Compliance Validation
Cryptographic Risk Scoring
Mission-Aware Cryptographic Risk Reporting
Executive Cryptographic Reporting
Continuous Monitoring
Enterprise Cryptographic Governance
```

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

```
Specification ID : IS-032
Title : AQELYN Secrets Security & Cryptographic Asset Intelligence Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0032
```

34. EA-0032 Engineering Objective

The objective of IS-032 was to introduce a dedicated Secrets Security & Cryptographic Asset Intelligence Engine that enables AQELYN to continuously discover secrets, inventory cryptographic assets, manage certificate lifecycle, analyze secret exposure, validate cryptographic compliance, score cryptographic risk, and generate remediation recommendations.

The engine extends AQELYN from data security posture management into cryptographic governance, certificate lifecycle intelligence, and enterprise secrets security.

35. EA-0032 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Secret discovery
- Cryptographic asset inventory
- Certificate lifecycle management
- Key management intelligence
- Secret exposure analysis
- Cryptographic compliance validation
- Cryptographic risk scoring
- Cryptographic remediation recommendations
- Knowledge Graph integration
- Security Data Lake integration
- Risk and threat intelligence integration
- AI Decision integration
- Executive reporting integration
- Event publishing

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Cryptographic Security Engine

Secrets security and cryptographic asset intelligence are implemented as a standalone engine rather than embedded in cloud posture, DSPM, or identity governance.

Rationale:

- Clear separation of cryptographic posture from data and identity posture.
- Independent lifecycle and governance.
- Better support for certificate lifecycle, key governance, and secret exposure analysis.
- Improved traceability for cryptographic assessments.

36.2 Decision 2 - Secrets and Cryptographic Assets Are First-Class Objects

Secrets, keys, certificates, and exposures are modeled as governed, evidence-backed objects.

Benefits:

- Cryptographic inventory becomes auditable.
- Secret exposure can be traced to assets and owners.
- Key and certificate lifecycle management becomes measurable.

36.3 Decision 3 - Certificate Lifecycle Is Continuous

Certificate expiration, renewal, validation, ownership, and chain integrity are monitored continuously.

Benefits:

- Expired certificates can be prevented.
- Service outages from certificate failures are reduced.
- Certificate compliance can be reported to executives.

36.4 Decision 4 - Key Governance Is Risk-Based

Key rotation, key age, algorithm validation, key ownership, and key usage analytics inform risk.

Benefits:

- Weak or stale keys can be prioritized.
- Algorithm policy violations can be detected.
- Mission-critical keys can be governed more strictly.

36.5 Decision 5 - Event-Driven Cryptographic Lifecycle

Secret discovery, certificate lifecycle, key rotation, exposure, policy, risk, and remediation events are published through the AQELYN Event Bus.

Examples include:

```
secret.discovered
secret.updated
secret.removed
certificate.expiring
certificate.expired
certificate.renewed
key.rotation.required
key.rotated
key.policy.violation
secret.exposure.detected
cryptography.policy.violation
cryptography.risk.updated
cryptography.remediation.recommended
```

36.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```
SecretAsset
CryptographicKey
CertificateAsset
CryptographicExposure
```

37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Secret, key, certificate, exposure objects |

| IS-003 Event Bus | Secret, certificate, key, exposure, risk events |

| IS-004 Evidence Engine | Evidence references and cryptographic support |

| IS-005 Knowledge Graph | Application, secret, key, certificate, exposure relationships |

| IS-006 Trust Engine | Data confidence and cryptographic trust |

| IS-007 Mission Engine | Mission-aware cryptographic risk prioritization |

| IS-008 Workflow Engine | Rotation and remediation workflows |

| IS-009 Policy Engine | Cryptographic policy validation and governance |

| IS-010 Compliance Engine | Certificate, key, and secret governance reporting |

| IS-011 Identity Governance Engine | Ownership, access, and identity context |

| IS-013 Risk Intelligence Engine | Risk scoring and business impact |

| IS-014 Threat Intelligence Engine | Secret exposure and key compromise context |

| IS-015 Security Operations Engine | Cryptographic incidents and SOC workflows |

| IS-019 Security Data Lake | Secret inventories, certificate metadata, key telemetry |

IS-020 AI Decision Engine	Remediation recommendations and confidence scoring
IS-022 Executive Reporting Engine	Executive cryptographic posture summaries
IS-028 Cloud Security Posture Engine	Cloud KMS, cloud secrets manager context
IS-031 DSPM Intelligence Engine	Data protection and data exposure context

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/cryptographic_security/  
■■■ tests/cryptographic_security/  
■■■ api/cryptographic_security/  
■■■ docs/cryptographic_security/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces cryptographic-security-specific controls:

```
Policy-governed cryptographic assessments  
Evidence-backed secret discovery  
Continuous certificate lifecycle tracking  
Key rotation governance  
Secret exposure detection  
Cryptographic compliance validation  
Cryptographic risk scoring  
Remediation ownership tracking  
Role-authorized cryptographic administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Continuous secret discovery  
Cryptographic asset inventory  
Certificate lifecycle management  
Encryption key governance  
Secret exposure analysis  
Cryptographic policy validation  
Mission-aware cryptographic risk reporting  
Executive cryptographic posture summaries  
Continuous reassessment  
Enterprise cryptographic governance
```

41. Risks Identified

| Risk | Mitigation |

|---|---|

| Undetected exposed secrets | Secret Discovery Engine and Secret Exposure Analysis Engine |

| Expired certificates | Certificate Lifecycle Engine |

| Weak algorithms | Cryptographic Compliance Engine |

| Stale keys | Key Management Intelligence Engine and rotation events |

| Missing ownership | Cryptographic Asset Inventory Engine |

| Unauthorized rotation | Policy enforcement and approval workflows |

| Poor auditability | Evidence-backed assessments and immutable audit events |

| Large vault scale | Distributed inventory and scalable telemetry ingestion |

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

- Unit testing
- Integration testing
- System testing
- Security testing
- Regression testing

Acceptance criteria cover secret discovery, cryptographic asset inventory, certificate lifecycle, key management intelligence, exposure analysis, cryptographic compliance, cryptographic risk, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

- Modular architecture
- Event-driven communication
- Immutable evidence references
- Traceability
- Explainability
- Security by design
- Repository stability
- Backward compatibility
- Governance-before-archive discipline

44. Dependencies

Required:

- EA-0001 through EA-0031
- IS-001 through IS-032

Enables:

- IS-033 and subsequent identity security posture, privacy intelligence, resilience, and enterprise security intelligence components

45. Completion Record

```

Engineering Archive : EA-0032
Implementation Specification : IS-032
Title : AQELYN Secrets Security & Cryptographic Asset Intelligence Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED

```

46. Archive Index Update

```

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
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EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
EA-0022 IS-022 AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023 IS-023 AQELYN Threat Exposure & Attack Surface Management Engine
EA-0024 IS-024 AQELYN Vulnerability Intelligence & Prioritization Engine
EA-0025 IS-025 AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
EA-0026 IS-026 AQELYN Configuration Compliance & Drift Intelligence Engine
EA-0027 IS-027 AQELYN Identity Threat Detection & Behavioral Analytics Engine
EA-0028 IS-028 AQELYN Cloud Security Posture Management Intelligence Engine
EA-0029 IS-029 AQELYN SaaS Security Posture Management Intelligence Engine
EA-0030 IS-030 AQELYN Software Supply Chain Security & SBOM Intelligence Engine
EA-0031 IS-031 AQELYN Data Security Posture Management Intelligence Engine
EA-0032 IS-032 AQELYN Secrets Security & Cryptographic Asset Intelligence Engine

```

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0032

Current Status:
EA-0032 COMPLETE

Next Implementation Specification:
IS-033 - AQELYN Identity Security Posture Management Intelligence Engine

48. Engineering Archive Publication Standard

EA-0032 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```

EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■

```

```

■■■ examples/
■ ■■■ example_*.md
■
■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■■■ images/
■
■■■ journal/
■ ■■■ Engineering_Journal.md
■
■■■ manifest/
■ ■■■ manifest.json
■
■■■ MD/
■ ■■■ EA-xxxx.md
■
■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-032-001	Continuously discover secrets	Sections 8, 12, 23	Complete
FR-032-002	Inventory cryptographic assets	Sections 8, 12, 23	Complete
FR-032-003	Detect cryptographic risks	Sections 8, 12, 23	Complete
FR-032-004	Provide explainable cryptographic intelligence	Sections 8, 22, 36	Complete
FR-032-005	Support cryptographic governance	Sections 8, 22, 23	Complete
FR-032-006	Publish cryptographic events	Sections 8, 15, 36	Complete
NFR-032-001	Continuous assessment	Sections 9, 24	Complete
NFR-032-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-032-003	Low-latency cryptographic analysis	Sections 9, 25	Complete
NFR-032-004	Explainability	Sections 9, 22, 36	Complete
NFR-032-005	Auditability	Sections 9, 27, 39	Complete
NFR-032-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source	Target	Relationship
IS-032 Purpose	EA-0032 Objective	Defines why the engine exists
Secret Discovery Engine	FR-032-001	Discovers secrets and credentials
Cryptographic Asset Inventory Engine	FR-032-002	Inventories keys, certificates, and cryptographic assets

| Certificate Lifecycle Engine | FR-032-003 | Tracks certificate expiration and renewal |

| Key Management Intelligence Engine | FR-032-003 | Evaluates rotation, age, ownership, and algorithm policy |

| Secret Exposure Analysis Engine | FR-032-003 | Detects exposed or hardcoded secrets |

| Cryptographic Compliance Engine | Governance and compliance | Maps cryptographic assets to policy |

| Cryptographic Risk Engine | Risk scoring | Calculates cryptographic security risk |

| Event Publisher | FR-032-006 | Publishes secret, certificate, key, exposure, and risk events |

| Security Data Lake Integration | Cryptographic telemetry | Supplies secret inventories, certificate metadata, key metadata |

| AI Decision Integration | Remediation recommendations | Supplies confidence and recommendations |

| Risk & Threat Integration | Threat context | Supplies threat, identity, supply chain, and DSPM context |

| Compliance Integration | Governance and audit | Supports certificate, key, and secret governance |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0032

EA-0032 was created to archive completion of IS-032 - AQELYN Secrets Security & Cryptographic Asset Intelligence Engine.

The archive records the expansion of AQELYN into secrets security and cryptographic asset intelligence. IS-032 defines the structure needed to discover secrets, inventory keys and certificates, monitor certificate lifecycle, evaluate key governance, detect secret exposure, validate cryptographic compliance, calculate cryptographic risk, generate remediation recommendations, and publish cryptographic security events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Cryptographic posture must be modeled separately from DSPM, cloud posture, and identity posture. DSPM focuses on data location and exposure; cryptographic security focuses on secrets, keys, certificates, algorithms, lifecycle, and trust.

Governance Note

EA-0032 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Secret Asset

```
secret_id: SEC-0001
secret_type: api_key
owner: platform_security_team
storage_location: vault://prod/payments/api-key
```

52.2 Example Cryptographic Key

```
key_id: KEY-2001
algorithm: AES-256
strength: strong
rotation_date: 2026-10-07
```

```
owner: data_security_team
```

52.3 Example Certificate Asset

```
certificate_id: CERT-3001  
issuer: enterprise_internal_ca  
expiration_date: 2026-08-15  
validation_status: valid
```

52.4 Example Cryptographic Event

```
{  
  "event_type": "certificate.expiring",  
  "certificate_id": "CERT-3001",  
  "days_until_expiration": 30,  
  "source_engine": "aqelyn_secrets_security_cryptographic_asset_intelligence_engine"  
}
```

53. Manifest Summary

Archive contents include:

```
README.md  
MD/EA-0032.md  
PDF/EA-0032.pdf  
HTML/index.html  
HTML/styles.css  
manifest/manifest.json  
requirements/Requirements_Matrix.md  
traceability/Traceability_Matrix.md  
journal/Engineering_Journal.md  
diagrams/Architecture.svg  
diagrams/Component.svg  
diagrams/Workflow.svg  
diagrams/EventFlow.svg  
diagrams/Integration.svg  
examples/example_cryptographic_security.md
```

54. Final Archive Statement

EA-0032 is the Engineering Archive for IS-032 - AQELYN Secrets Security & Cryptographic Asset Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0032 COMPLETE  
IS-032 COMPLETE  
NEXT: IS-033
```